

- 2 Mercerisation is a treatment used to increase the tensile strength of natural plant fibres. In this process, fibres are treated with sodium hydroxide solution.

A study was carried out to determine the concentration of sodium hydroxide solution to produce the strongest fibres.

Samples of fibres were obtained from four Nigerian plants: baobab, roselle, okra and kenaf.

All of these fibres were then mercerised by placing them in four different concentrations of sodium hydroxide solution.

The tensile strength of these fibres was then measured.

- (a) (i) State the independent variable in this investigation.

(1)

- (ii) State how one named variable could be controlled when mercerising the fibres.

(2)

Variable

How the variable could be controlled

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- (iii) Devise a method that can be used to make a valid comparison of the tensile strength of fibres from the same plant, treated with different concentrations of sodium hydroxide solution.

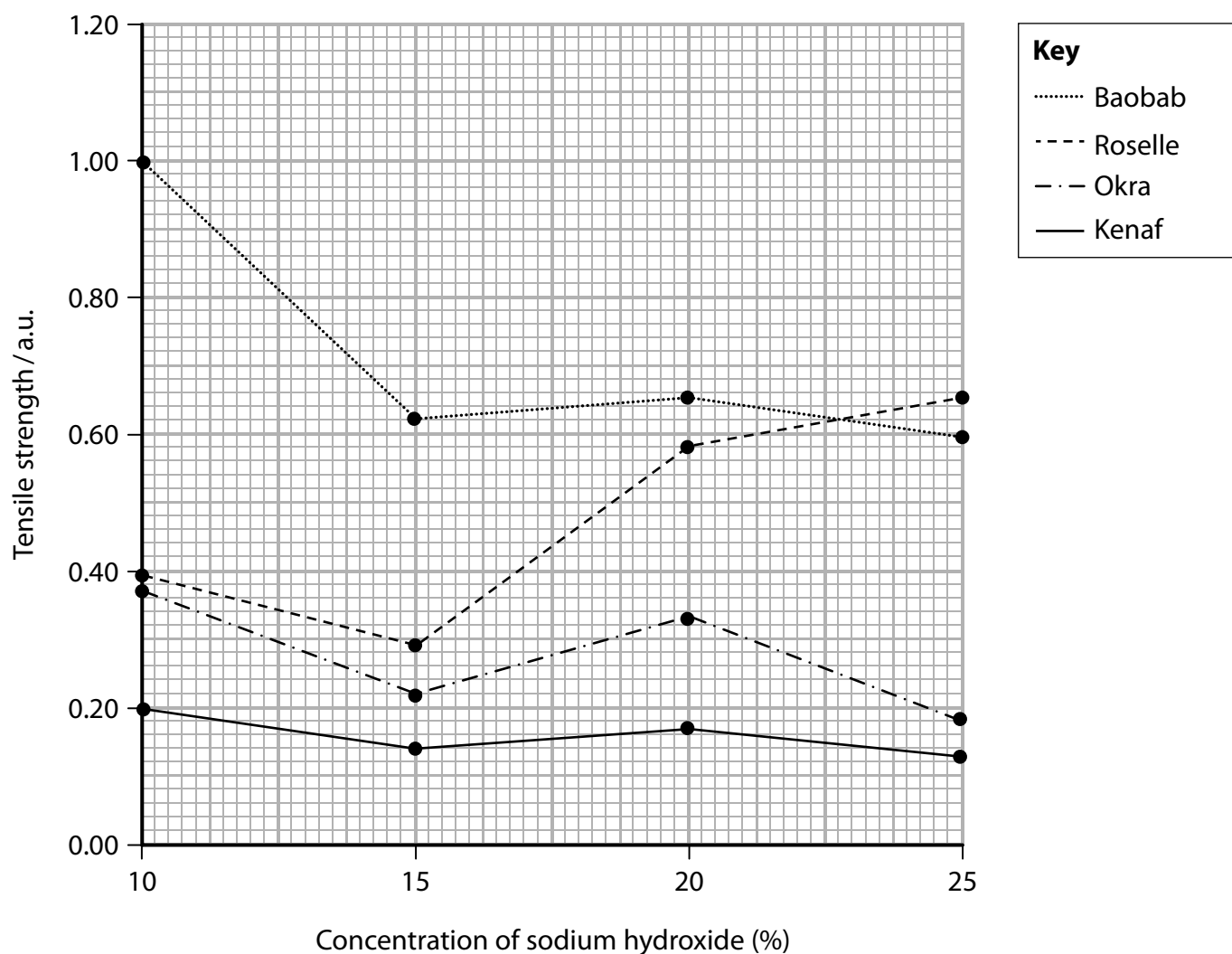
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(b) The graph shows the tensile strength of fibres from these four plants, treated with sodium hydroxide solutions.



(i) Using the graph, determine the concentration of sodium hydroxide solution that should be used for mercerisation of fibres from each of these four plants.

(2)

(ii) Draw a suitable table to include the results for roselle and baobab only.

(3)

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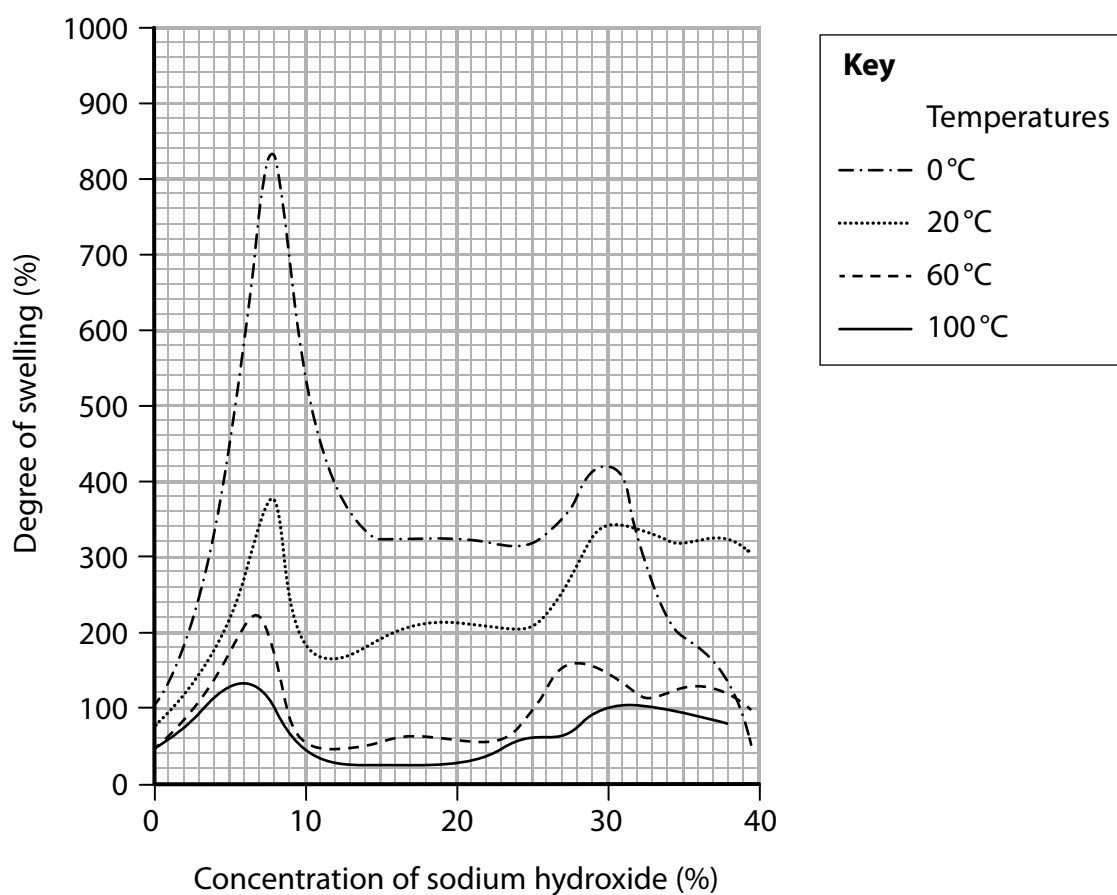
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(c) Mercerisation causes plant fibres to swell.

A study was carried out on the effect of temperature and the concentration of sodium hydroxide on the degree of swelling of cotton fibres.

The graph shows the results of this study.



(i) Compare and contrast the trends shown by the results at 20°C with those at 0°C.

(2)

- (ii) Describe the effect of temperature on the degree of swelling of cotton fibres at a sodium hydroxide concentration of 5%.

(2)

(Total for Question 2 = 17 marks)